



Regular Article

Linkages between Traditional Water Systems (TWS) and Sustainable Development Goals (SDGs): A case of Govardhan, India

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1. Introduction

Water, being vital for life, development, and the environment, is a precious natural resource, and its availability and management can have significant consequences on humans and the environment (UNWA, 2003, pp. 1–34). Water plays a crucial role in the growth and development of cities and communities. The traditional knowledge, culture, and practices surrounding water management are deeply rooted in the history of human civilisation, as evidenced by the fact that many ancient societies thrived near rivers and other bodies of water (Sadhale, 2006), (Selvaraj et al., 2022). Historically, there appears to be a relationship between the increased human efforts to construct water harvesting structures and sudden climate changes such as dry spells, droughts, and floods (Pandey et al., 2003). Communities with longstanding ties to their ancestral lands, often called Indigenous Peoples, have held authority over the water and land within their territories for countless generations (Nicole Wilson et al., 2018). This correlation can be observed across different regions, where communities have had to adapt to a changing climate by building water harvesting structures, like dams, reservoirs, canals, and cisterns, to secure access to water. For example, during drought, communities would build water harvesting structures to collect and store rainwater, thus ensuring a reliable water source. Similarly, during floods, communities would build water harvesting structures to protect against flood damage and direct excess water to areas where it could be stored or used beneficially. This adaptation to climate fluctuations through the construction of water harvesting structures has been a common practice throughout history. It has been a key factor in the survival of many ancient societies. People have often settled along rivers, lakes, and coastlines to guarantee access to vital water resources. In arid to semi-arid regions, which face scarcity of water owing to environmental conditions and highly varying rainfall patterns, people have developed several immensely rich and varied, often interconnected, systems to manage water that has proved to be sustainable over centuries (Kzonig, 2001), (Ramineni & Bharadwaj, 2021). Today,

these traditional water systems are considered heritage, a term used broadly to encompass both recognized World Heritage Sites and the historic built environment that people have chosen to preserve. These systems, that range from small-scale water harvesting systems and aqueducts to larger water pumping facilities, irrigation and drainage networks, dike systems, and defence systems, not only have historical and cultural significance but also continue to be functional and provide a reliable source of water. They are a valuable source of knowledge and inspiration for addressing current and future water management challenges (Hein, 2019). All traditions across the world have a special reverence for water, which has led to the patronage of building and managing water bodies by the communities (Mishra, 2011).

Climate change is causing changes in precipitation patterns and sea level rise, which in turn affects water resources and heritage. Pollution also affects them, making the water unsafe for consumption and damaging traditional systems. Additionally, changing political and societal patterns can also have an impact on water resources, as development and urbanization can lead to the destruction of traditional water systems. The effects of global climate change, pollution, and shifting political and societal patterns are felt on various levels. These effects include changes to systems for drinking water, irrigation, and drainage, as well as the systems of coastal areas, deltas, and port cities (Garot & Garot, 2014; Okamura, 2011; Willems & Van Schaik, 2017). In today's world, access to water is a significant constraint for both agricultural production and income (Namara et al., 2010). Water plays an important role in the world economy and sustainable development. Poverty reduction, food security, job creation, and GDP growth are all closely linked to access to clean water and adequate water resources. Water is an essential element in the provision of food security, the creation of jobs, and the growth of the global economy (Bhattacharya, 2015), (Machiwal et al., 2018). Climate change is having a significant impact on the environment and societies worldwide, and to address it, many adaptation strategies, particularly in the area of water management, are becoming increasingly important (Dean et al., 2016). Sustainable water

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management is vital to meet the needs of the current and future populations while preserving the natural environment and biodiversity. It requires balancing the competing water demands, protecting, and restoring water systems, and ensuring that all people have access to clean and safe water (Selvaraj et al., 2022).

According to the Our Common Future report (1987) prepared by the Brundtland Commission (Our Common Future), sustainable development is referred to as "development which meets the needs of the present without compromising the ability of future generations to meet their own needs". On June 1992, at the Earth Summit in Brazil, more than 178 countries adopted Agenda 21, a comprehensive plan of action to build a global partnership for sustainable development to improve human lives and protect the environment. Member States unanimously adopted the Millennium Declaration at the Millennium Summit in September 2000 (Turcea & Ion, 2020). The Summit led to the elaboration of eight Millennium Development Goals (MDGs) to reduce extreme poverty by 2015 (Klarin, 2018). Goals were intended to Eradicate extreme poverty and hunger, achieve education for all, gender equality, Reduction in child mortality, improve maternal health and combat deadly diseases like HIV/AIDS and malaria and other diseases, environmental sustainability and develop a global partnership for development (Churchill, 2020). MDGs had various successes and failures (Kumar et al., 2016). There were reasons that led to their discontinuation were the thematic and regional unbalance as many countries adopted a "piecemeal approach" (Global Taskforce, 2016). Despite their failure, it built a foundation for New sustainable development goals as they spearheaded an unprecedented international movement against extreme poverty, reducing it by more than 50 percent globally and a coherent catalogue of goals at the global level and provided a holistic approach to the individual thematic areas like eliminating income poverty and promoting literacy etc. (Kumar et al., 2016). In 2015 as a successor to the Millennium Development Goals (MDGs) (Woodbridge, 2015), the United Nations (UN) adopted the 2030 Agenda for Sustainable Development. It includes 17 Sustainable Development Goals (SDGs), 169 integrated and indivisible targets, and 231 unique indicators (U. & W. B. Group, 2016). SDGs are planned for a sustainable society where social inclusion, environmental sustainability, and economic progress are valued equally (Goubran & Cucuzzella, 2024). According to the 2030 agenda, the 17 goals are directly or indirectly interlinked with each other. The goals are designed in line with the socio-economic and environmental dimensions of sustainable development. The Agenda 21 of the Rio Earth Summit in 1992 (De, 1993) where the concept of sustainable development emerged, recognized the vital role of Indigenous People and other local communities in the management of traditional water systems. This is due to their traditional knowledge and associated practices that have been developed over generations in response to the unique environmental conditions of their regions (Kimerling, 2002). Indigenous People and local communities often have a deep understanding of traditional water systems, closely associated with territory, including techniques for harvesting, storing, and distributing water. This knowledge and practices are essential in maintaining the ecological balance and biodiversity of a region, preserving cultural and natural heritage sites, and adapting to changing environmental conditions.

This study aims to identify the linkages between the Traditional Water Systems (TWS) and the relevant SDGs. It inquires whether there are any goals and targets related to TWS and whether TWS can help in achieving them. Understanding the linkages between traditional water systems and sustainable development goals is important to conserve natural resources, protect ecosystems, empower communities, adapt to changing climate, and preserve cultural heritage. It's crucial to consider the interconnections between water resources, heritage, sustainable development and others when designing policies and implementing a project.

2. Methodology

The methodology has two major stages; first to develop an understanding of the traditional water system of Govardhan, India; second to identify linkages with the goals and targets of SDGs. The study is based on a multi-disciplinary approach that combines existing research on the Kunds of Govardhan (Sharma et al., 2021), (Sharma, 2021); a comprehensive literature review (Snyder, 2019) to gain a thorough understanding of the linkages between TWS and SDGs, field study, and data analysis.

The first stage began with data collection and documentation of the built and natural components in the study area. The primary and secondary surveys, literature studies, and interviews with various stakeholders were used to gather the necessary data for the study. The mythological and cultural associations of Kunds in terms of Govardhan, Braj, and the larger Krishna mythos, which also have a ritual-based significance, were identified through religious texts and primary data collection. Scaled drawings were prepared based on primary observation, by integrating site measurements with GIS mapping. The geographical character and watershed delineation was firstly done through secondary data collection using books and reports by the Central Ground Water Board (CGWB), The Ministry of Environment, Forest and Climate Change (MoEF&CC) etc., and using digital tools like GIS, Bhuvan, etc. The gaps in data were identified and resolved by primary data collection and site visits. The research also focused on community-participatory traditional water management systems developed over a period through local knowledge sources by conducting interviews with community members and experts. The perspective through which Kunds were observed and studied is broadly divided into three heads - architecture, water, and geography. Kunds were visualized as a system of multiple elements coming together - as trends observed on-site, the conservation of built fabric is done extensively, but the ecological and cultural continuum of practices is not taken into consideration.

In the second stage, we conducted an extensive literature survey to find out papers and reports related to SDGs. Based on the findings from the first stage and the extensive literature review related to SDGs. To map the synergies and trade-offs between TWS and SDGs, we have identified specific SDGs goals that traditional water systems can contribute to, such as SDG 6: Clean Water and Sanitation, SDG 11: Sustainable Cities and Communities, SDG 14: Life Below Water, SDG 15: Life on Land, SDG 16: Peace, Justice and Strong Institutions, and SDG 17: Partnerships for the Goals. By highlighting these linkages, we have demonstrated the importance of TWS in achieving sustainable development.

3. Traditional water systems of India: Overview

Traditional water conservation and management techniques exist in every region of India (Murthy et al., 2022). For instance, groundwater recharging has been practised traditionally in parts of Delhi, Gujarat, and Rajasthan, because of low rainfall and extreme weather patterns. Water systems were developed specifically to store water in regions where runoff is relatively higher and more widely distributed. Traditional water management knowledge is based on different usage trends, the soil state of the field, the rainfall pattern of the city, and inter-surface soil fractures. Communities used to choose rainwater harvesting sites based on these characteristics, by first planning an outline of anticipated rainwater runoff flow slopes or directions, and then building drainage outlets for the region's excess water to evacuate. The outlines, elongation, height, and breadth measurements were sketched out to elevate the barriers laterally opposing the water flow-in ditches (Bhaduri & Singh, 2012, pp. 1–19). The material excavated was used to construct the fences. The hydro-experts preferred this approach because it was the easiest (Husain & Arif, 2014). These systems are still vital and functional in many places and are now under threat with the thrust of urbanization. Traditional knowledge of water harvesting and management in various

forms has developed in India based on various processes of the hydrological cycle, evaporation, condensation, rainfall, streamflow, physiography, etc. Religious associations not only provided cultural significance but also long-term protection. Depending on ecological differences, many trends of rainwater harvesting systems in India have been developed. The northwestern Himalayas make up India's high-land ecological zone, where rain harvesting systems like *Zings* and *Kuls* are often used. *Kuls* are small tanks used to collect melting glacier water, whereas *Zings* are used to divert water from glaciers to settlements (More, 2020). India's Arid ecological zone includes the southwestern parts of the Haryana and Punjab states, the Kacchh peninsula, the western parts of Rajasthan, and the northern portion of Gujarat state's Kathiawar peninsula. Rainwater harvesting systems such as *Kuis* and *dakeriyan* are used in the western part of Rajasthan. *Kuis* are subterranean 10-12-m-deep tanks used for water storage. Irrigation techniques such as *Khadins/Dhora* are also used in Western Rajasthan (Hussain et al., 2014). The northern plains, Gujarat, the central highlands, and the Deccan plateau make up the semi-arid ecological area, where *viridas* are practised in low depressions, which are shallow wells. For groundwater recharge, other rainwater irrigation methods are used, such as *Johad*, *Baandh*, *Nada* (Kavarana). In this series of semi-arid traditional water systems, Kunds are the source of sustenance for the people of Braj and are also symbols of religious significance (Saha et al., 2010).

4. Traditional water systems of Govardhan

The cultural landscape of Govardhan, India is a unique combination of cultural, ecological, and religious elements. The central feature of this landscape is the Govardhan Hill, a natural geographic feature that changes with the changing climate. Other components of this system include forests (*Vans*), gardens (*Upvans*), water bodies (kund), and settlements, which are interconnected and form a single system. These elements are deeply connected and play a significant role in the religious beliefs of the local community, as they are considered sacred and associated with the early years of Lord Krishna's (Hindu god) life. The circumambulation (*parikrama*) of Govardhan Hill and Radha Kund is a sacred dynamic loop that holds all these components together. This cultural landscape is a result of *Vaishnavites*' (followers of God Vishnu or Krishna) designating particular sites for numerous incidents in the life of Lord Krishna and creating the countryside from memory and religious scriptures where he performed many leelas (Thakur, 2011). It is an associative cultural landscape that has been developed over time through human-nature interactions (Kanitkar).

5. Study area

The Cultural Landscape of Govardhan is in Braj region in northern India. The region is known for its association with Lord Krishna and is considered a major pilgrimage site for Hindus. It is characterized by a mix of flat plains, rolling hills, and river valleys. The Braj region is also known for its rich cultural heritage, including music, dance, and art, which reflect the traditions and beliefs of the local people (Sinha, 2014). The region is primarily located in the state of Uttar Pradesh, with parts extending into the neighboring state of Rajasthan and Haryana. While all the locations in Braj are associated with Lord Krishna, each of them has its own unique identity and significance. Each of these locations has its own distinct ecological, cultural and religious practices, which have evolved over time and are an important part of the local identity. But still considered a singular region because of their shared cultural and religious heritage. The people of Braj share a common language, customs, and traditions, which have been passed down through generations and have helped to create a unique cultural identity for the region. Some of the major locations and sites in Braj include Mathura, Vrindavan, Govardhan, Barsana, Nandgaon (Fig. 1).

Traditional water bodies are an example of such a component which can be found in almost every important site. For example, in Mathura,



Fig. 1. Important sites and locations in the Braj region.

there are several kunds such as Potra Kund, Kankhal Kund, and Balbhadr Kund, which are associated with various legends related to Lord Krishna's childhood. In Vrindavan, there are several kunds such as Brham kund, Shyam Kund, and Govind Kund, which are associated with Krishna's love for Radha. Similarly, in Govardhan, there are several kunds such as Manasi Ganga, Radha Kund, and Kusum Sarovar, which are associated with various legends related to Lord Krishna's life. In Nandgaon, there is a kund called Pavan Sarovar, which is associated with Lord Krishna's childhood. So, while the significance and legends associated with each kund may vary, they are a common feature across all the important sites in Braj and are an important part of the religious and cultural landscape of the region.

This study will be focusing on The cultural landscape of Govardhan sprawls around the Govardhan hill (also known as "Giriraj Parvat") and Radha Kund, located in the Mathura district of Uttar Pradesh and a small fraction in the Bharatpur district of Rajasthan. The study area is in the Govardhan and Mathura tehsil in the Mathura district of U.P and in the Deeg tehsil of Bharatpur district of Rajasthan (Fig. 2). The study area encompasses 7 villages, including Poonchari, Anyore, Jatipura, Sakitara, Govardhan Brahman, Govardhan Gorwan, Radha Kund Rural,

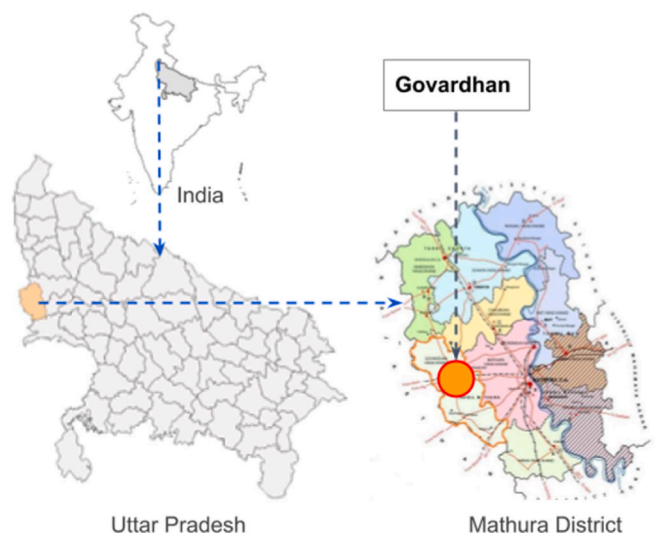


Fig. 2. Study Area lies in Govardhan Tehsil of Mathura district in Uttar Pradesh, India.

Govardhan and Radha Kund Nagar Panchayat. According to census data, the area has a population of 46,000 (of [Census & Operations, 2011](#), p. 590). However, being a prime centre for lord Krishna worship, Govardhan is visited by approximately 1 crore pilgrims annually ([UP Department of Tourism, 2020](#)). This high volume of visitors puts a strain on the area's resources, particularly water, which must be managed effectively in order to meet the needs of both the resident population and the influx of pilgrims.

The Govardhan hill and associated kunds are a natural-built form, whose origins can be found in the mythological interpretations of the Vaishnavite legend, which is central to the Indic culture, where Lord Krishna takes the centre stage in his various forms as the protagonist. The mythic stories of the conflict between Indra and Lord Krishna implored the people of Govardhan village to look towards spiritual means of worship, rather than sacrificial or materiality of worship. The image of lord Krishna lifting this very hill, with the tip of his finger, to shield the villagers and animals from the wrath of Indra in the form of rain and lightning, is of large significance in the Braj Mandal and the larger Indic culture.

5.1. The ecological context of Govardhan

Govardhan is a part of the western fringe of the Ganga alluvial plain. It lies in an older alluvium plain geomorphological unit. This unit is mainly characterized by yellow clay, *kankar* and *reh*. It is represented by Alluvium plain, Salt affected plain and waterlogged plain. Here slopes are very high and they mostly form run-off zones ([Gupta and Misra, 2019](#)). Govardhan lies in Yamuna's middle sub-basin and the slope of the land is towards the Yamuna from north to south and southeast direction. Govardhan lies on an average height of 180 m and the Yamuna at 150 m. Many seasonal streams pass and originate from Govardhan, which eventually gets connected to the Yamuna River.

The study area lies at 27°27'22.0"N 77°25'40.7"E to 27°31'56.7"N 77°29'55.0"E. Govardhan hill is a long, low ridge, of an ancient quartzite part of the Aravalli Mountain range ([Fig. 3](#)) and is the largest in this region. It looks like a huge collection of small rocks due to the metamorphosis of the sandstone and extends altogether for about 8.1 km ([Fig. 4](#)), rising up to 30 m above the surrounding plain ([Fig. 5](#)) ([Sinha, 2015](#)). The Aravalli, as a geological feature, is a rain-catcher, directing the rain-laden winds from the Western Ghats towards the Govardhan, contributing more to its ecological, agricultural, and culturally contributing phenomenon. Undulating topography around the hill allows rainwater to get collected into natural depressions, resulting in ponds, lakes, and wetlands formation. These also act as natural groundwater recharge zones. These natural phenomena make this region suitable for water-centric cultural development and it has been so since ancient times, taking place in mythology and history as well.

The unique alignment of the landscape, coupled with abundant rainfall and a network of rivers, directs rainwater runoff towards the

Yamuna River, leading to the formation of numerous seasonal streams and channels throughout the area. With time some of these streams are lined and converted into small canals ([Gupta and Misra, 2019](#)). It is observed that these channels also act as sources of water for many of the kunds. At times when water overflows from the kunds, especially during the rainy season, water gets released into these channels. Some kunds are not linked to any channel and instead depend on much larger catchment areas. Many orders of streams are found in the catchment areas around Govardhan hill, from where the surface runoff is carried to the kunds. The map below shows the unique interconnected system of water flow and collection ([Fig. 6](#)). It's worth noting that the presence of these natural hydrological systems has facilitated various human interventions and enhancements. These interventions have been made possible due to the region's inherent water resources and have further enriched the cultural significance of Govardhan.

5.2. The cultural context of Govardhan

Govardhan is one the most significant centre of Krishna Bhakti and Vaishnavism. Braj culture is mostly a forest culture. Numerous transcendental lush groves and hills on both sides of the Yamuna are mentioned in the Braj Bhakti Vilasa ([Gosvami, 2018](#)). According to the Shrimad Bhagavat Gita, Braj is Lord Krishna's birth and childhood place.

Govardhan is one of the most recognizable features in Krishna mythology, wherein Krishna lifts the hill to shield an entire village from the wrath of the rain god Indra. This eventually become a part of Hindu mythology as well ([Entwistle, 1987](#)). Braj which has been interpreted in many forms, like in Mathura Mandal ([Fig. 8](#)) and in the shape of a lotus or the *Braj-Kshetra* ([Fig. 9](#)), and the Govardhan are believed to be in the very heart of the sacred land of Braj ([Singh and Dubey, 1988](#)).

This region has been home to a diverse population of local inhabitants for thousands of years, each contributing to the area's rich socio-cultural fabric. The Jats, Gujars, Meos pastoral Abhiras (commonly referred as Ahirs) inhabited and roamed the region since time immemorable.

6. Traditional water system

Traditional knowledge pertinent to kunds and ancient water management systems in Govardhan was tacit knowledge and had to be decoded from the existing architecture and practices. An understanding of this knowledge was developed as a result of the field survey and interviews with experts and residents of the place. Glimpses of this system have been prepared upon interpreting the collective consciousness of the people.

6.1. Macro level

To understand and conserve these kunds understanding of Geography and Gravity is required, broadly there are two types of streams in this area. Stream A represents the flow of water from the top of the hill, eventually forming smaller streams that flow away from the feature. Stream B represents the flow of the water from sources other than the hill, that flows through and around the Govardhan ([Fig. 7](#)). There are three major sources of water to feed the kunds, incl. water from the catchment areas, subterranean water, and canals or channels.

6.2. Meso level

Based on the topography and nature of rainwater runoff, the design of the kund also changes. Two types of water systems have been developed, depending on the patterns of the stream flows.

6.2.1. Type A

This type of water system is found in the kunds near higher terrain or built near the hill. A majority of these kunds are used for both religious



Fig. 3. Govardhan Hill view.

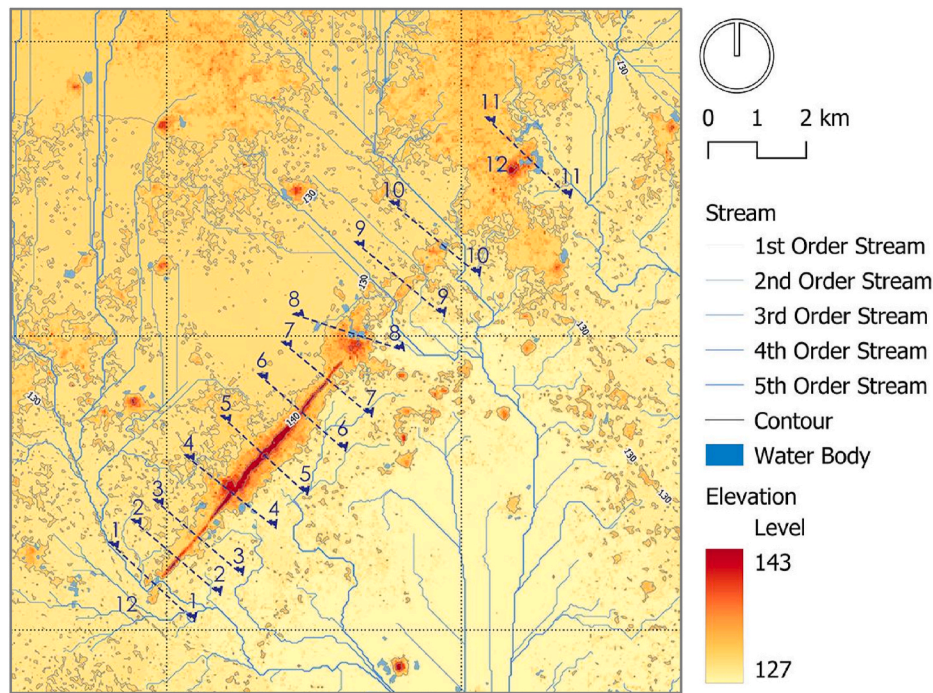


Fig. 4. Elevation map Govardhan (Sharma et al., 2021).

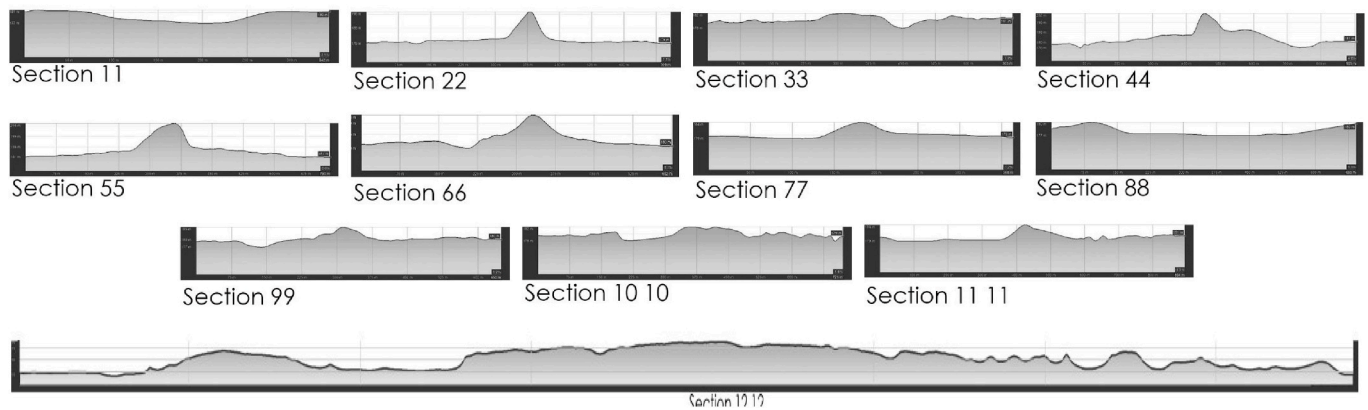


Fig. 5. Sections of the Govardhan Hill (Sharma et al., 2021).

as well as utility purposes and are also designed to facilitate groundwater recharge (Fig. 8). At the time of rainfall, water from the catchment gets collected in the kunds, leading to groundwater recharge in this area. The water overflowing from these kunds exits to the *Pokhara* towards a lower elevation. During the dry season when the water level decreases in the kunds, the water quality deteriorates due to an increase in micro-organisms and silt deposition, making it appear slimy. As a security measure, the recharged water can then be extracted from wells adjacent to kund, making them a critical component of these water systems. In Fig. 9 red dotted line represents the pre-monsoon underground water level while the blue line represents the post-monsoon level. Within the study area, the *Apsra Kund*, *Naval Kund*, *Hariju Kund*, *Rudra Kund*, *Sankarshan Kund*, and *Govind Kund* represents this category of kunds.

6.2.2. Type B

The water systems in the kunds which are away from the hill and spread along the Parikrama Marg till the Radha Kund are fed by the water streams formed on flatlands during rain, canals, and subterranean water. This particular type of system is mostly seen where post-rainfall rainwater gushes into the streams and channels that fill the kunds.

The size of these water systems, in terms of the number of water bodies connected to each other as well as the size of the kunds, is comparatively larger. Within the study area, rainwater collected in the water bodies situated on the northern offshores of the Govardhan flows towards the kunds. The water further flows towards lower areas on the southern side. This creates an interconnected network of water bodies within the area (Figs. 10 and 11). Radha Kund, Kusum Sarovar, Narad Kund, Uddhav Kund, Mansi Ganga and Ratna Kund are of this type.

6.3. Micro level

Each water system is composed of a number of different kinds of kunds, connected to each other through the existing channels. Even within a system, these kunds vary in terms of their usage and size, which practically determines their architectural design. As its definition states, a kund is a stepped tank of a particular architectural design and can also be looked like an expression of a cultural landscape. In Govardhan, kunds are like architectural elements designed in balance with nature, which not only allow human interaction with ecology but also act as an interface to connect with the divine (Fig. 12). Kunds also facilitate

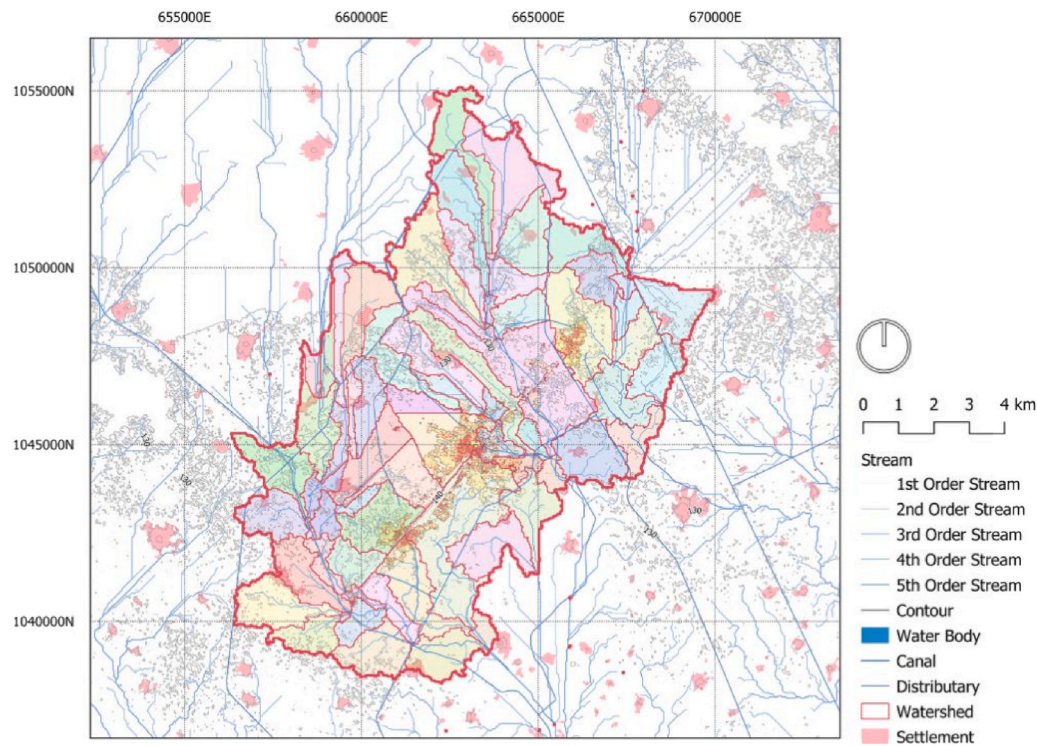


Fig. 6. Map showing the watershed of the Govardhan (Sharma et al., 2021).

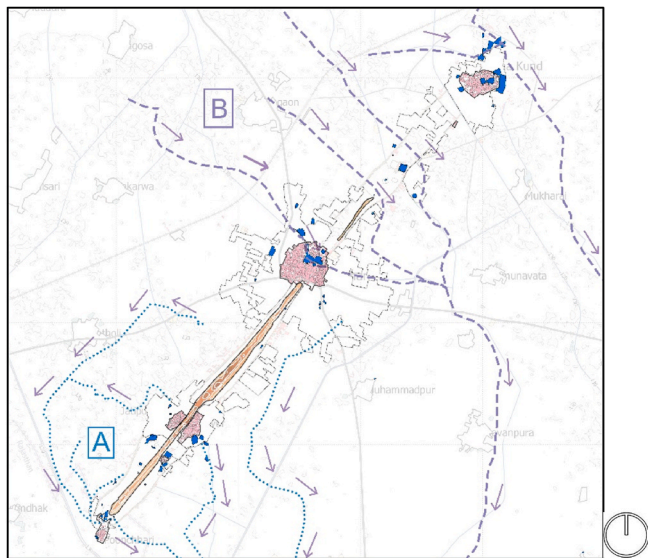


Fig. 7. Types of streams in Govardhan.
Source: STRM and OSM. Modified by Author.

essential community services like the provision of drinking water, rainwater harvesting and flood control. Kunds have played an essential role in the growth of settlements since the Vedic era, resulting in their inclusion in the mythical Vaishnavite narrative that took form in the 15th Century.

Over the years, kunds have proven essential for the sustenance of the living systems of the towns, making their place in the architectural vocabulary of the Mughal, Rajput and Jat Era of the region, and becoming instigators of artistic and creative forms of expression (Entwistle, 1987). The location of kunds around the Govardhan hill and parikrama indicate their direct association with the mythological and religious practices.

Being an essential component of the social infrastructure, these have

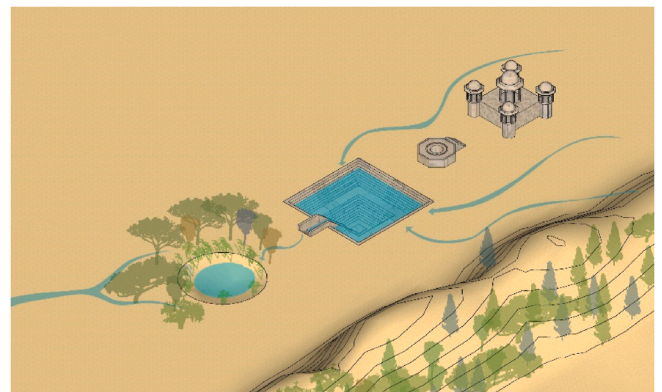


Fig. 8. Schematic plan of type-a stream water system.
Source: Author

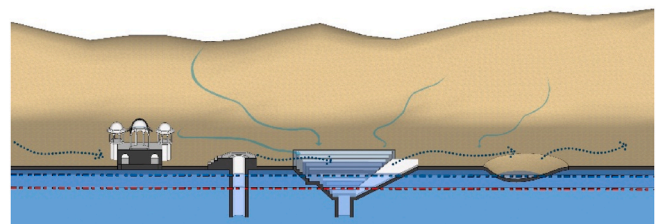


Fig. 9. Schematic Section type A stream water system.
Source: Author

become places of social gathering and interaction among the local and pilgrim communities. With their multiple built and unbuilt components, kunds have become an integral part of the ecosystem. This support creating an environment that is in equilibrium and creating a stage for a unique ecological setting for the flora, fauna and human growth. The

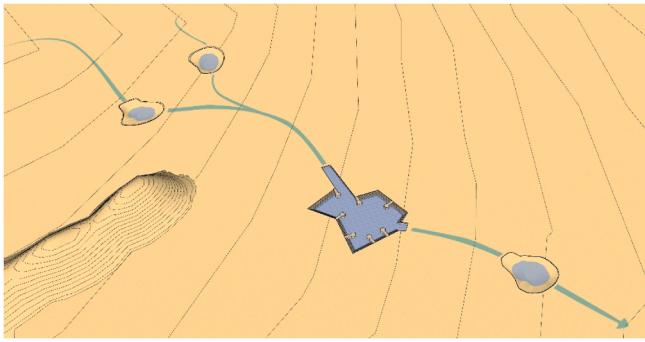


Fig. 10. Schematic plan of type B stream water system.
Source: Author

extensive use of these kunds over the ages reflects upon their religious, social and infrastructural significance in sustaining the population, cultural relationships and artistic imagery within the Govardhan region. At present, there are around 49 kunds of different shapes and sizes, in and around the 7 kos (mile) or 21 km parikrama of Govardhan hill and Radha Kund (Fig. 13).

6.3.1. Identified architectural component of kunds

These kunds are built out of red sandstone and locally backed lakhori bricks, which are abundantly available in the adjoining areas of Rajasthan. Some of the tanks are built using boulders all around. These kunds are generally used for drinking, bathing and cattle use.

The different components found in the kund of Govardhan along

with their schematic, are detailed in the Table 1 below.

Kunds are an important part of the socio-cultural fabric of Govardhan, and essential for the local communities and the pilgrims visiting the area. Activities such as bathing, religious rituals, social and religious gatherings, domestic usage of water, washing, etc. are heavily dependent on the local kunds. These kunds also play a significant role in the social fabric of any place, by becoming catalysts for communication and recreation, serving as activity spaces in a locality.

7. Discussion

During this research, an attempt was made to comprehend and document Govardhan's traditional water systems. The Kunds of Govardhan are an important part of the ecological setting of the Yamuna Basin. Govardhan itself is offshore of the Aravalli Ranges, making it a significant setting for restoring the crucial balance between nature and settlements. These kunds are a part of the larger water system in Govardhan and have been serving as an essential component for sustaining large settlements, especially in such semi-arid climates. The kunds are a part of the mythic narratives of Braj and Govardhan, as an archetypal setting of imagined cultural landscapes. Kunds are also integrated into the routes of cultural and religious parikramas and ensure the continuum of these parikramas. These systems help in the sustenance of a large number of people, flora and fauna, and their associated living systems.

These water systems can be visualized as a system of multiple elements coming together - as trends have been seen on site, the conservation of built fabric is done extensively, but the ecological and cultural continuum of practices is not taken into consideration. Water, its

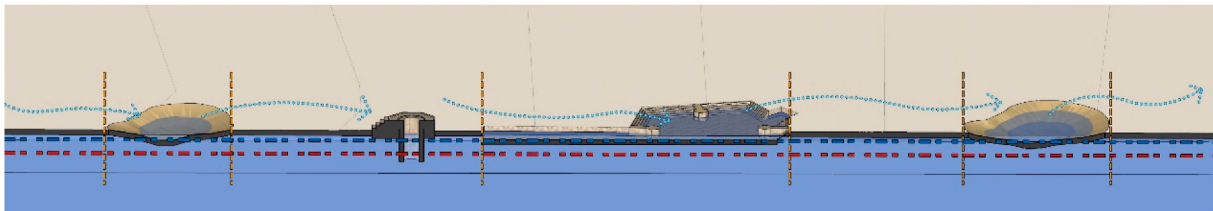


Fig. 11. Schematic Section type B stream water system.
Source: Author

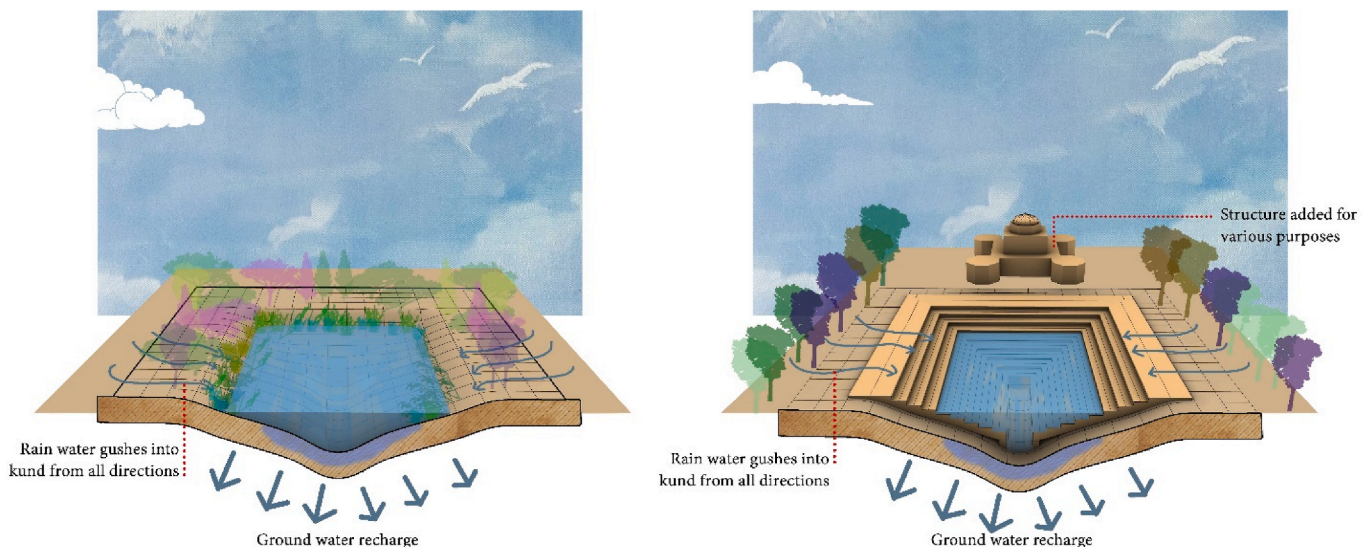


Fig. 12. Typical Kund design approach adopted in this region (a) In a natural setting where rainwater is getting collected into the depression, with naturally occurring flora and fauna. (b) Designed architectural elements creating space for human interaction with the ecosystem.

Source: Author

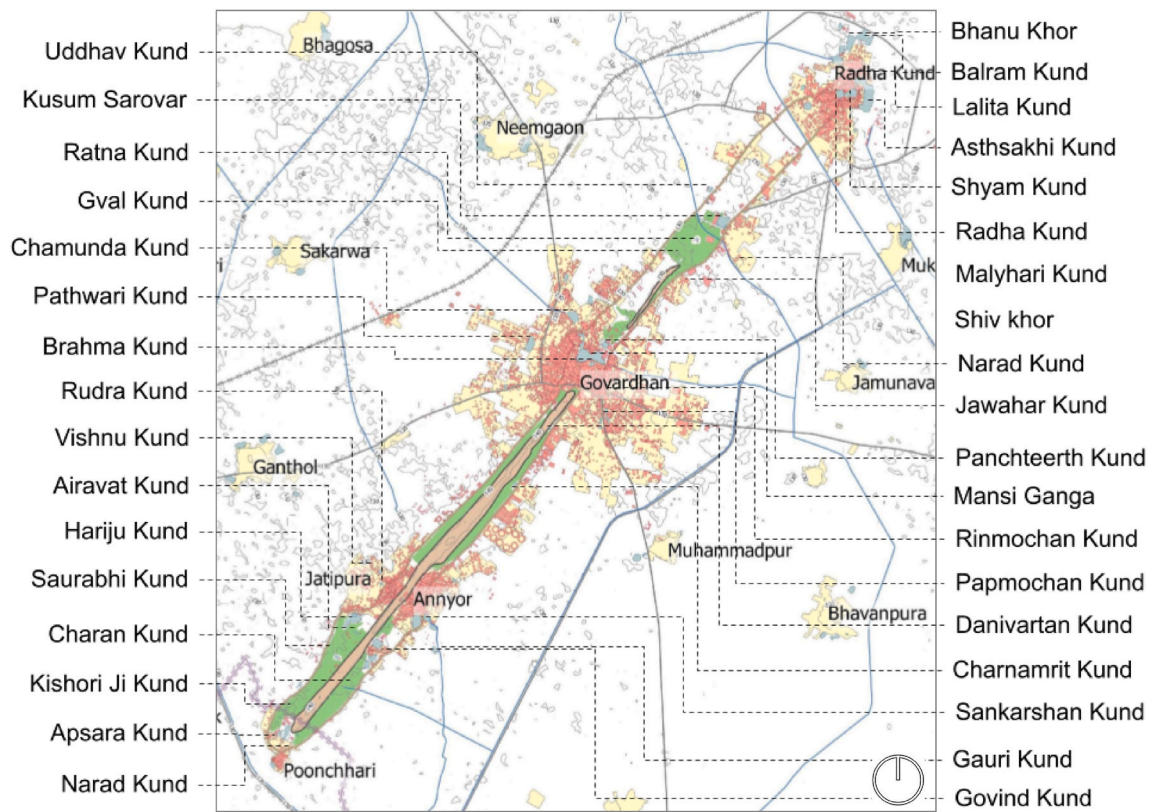


Fig. 13. Kunds sprawled in the Cultural Landscape of Govardhan.

Source: STRM and OSM. Redrawn by: Author

quality, cultural practice, architecture (Fig. 14) and these aspects come together to form a kund.

As in many other regions, local communities, who have historically lived in the area, practiced unique traditions, and maintained a distinct cultural identity. However, these communities may not always receive the recognition and rights they deserve, like how Indigenous Peoples globally have faced struggles in asserting their rights (Bhagabati, 2023). While the term "Indigenous" can be complex and debated in the Indian context, the Jats, Gujars, Ahirs or Abhiras are among the communities with a deep-rooted presence in the Braj region, which includes places like Govardhan.

Jats, the group with various subcastes, holding significant land-ownership and playing a central role in agriculture, in Braj region at large. Alongside them, the Gujars, historically engaged in cultivation and cattle-keeping, have left their mark, although many migrated from the Braj region due to historical events. In the northwestern reaches of Braj, the Meos form a distinct community, blending Muslim and Hindu customs. Scheduled castes, notably the Chamars or Jatavs, constitute a crucial agricultural labor force, contributing significantly to the region's economy. The Abhiras, believed to have roamed Braj in ancient times and associated with cow herding, add to the area's rich cultural heritage.

Abhiras, who were pastoral tribe believed to have roamed around the Braj region in early periods. They are associated with cow herding and are sometimes referred to as Ahirs in modern times. Over the centuries, they adopted certain aspects of the culture, beliefs, and rituals associated with the higher-status Kshatriya caste. This included claiming descent from the Yadu dynasty to which Lord Krishna belonged and emphasizing their connection to the Kshatriya warrior class. By doing so, they aimed to elevate their social status and gain acceptance within the broader Hindu society (Entwistle, 1987). As this process of Sanskritisation continued, the Abhiras began to identify themselves as Yadavs, a term associated with the Yadu dynasty, and adopted the cultural practices and lifestyle elements associated with the Yadavs, such as those of

herdsmen, cowherds, and milk sellers (Jaffrelot, 2003). Over time, this transformation became more pronounced, and the Abhiras gradually integrated into the Yadav community, which encompasses various castes across different regions of India.

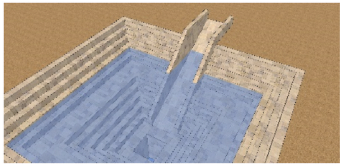
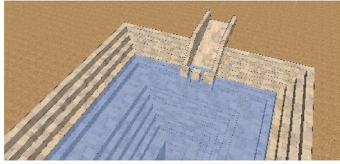
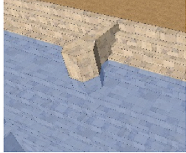
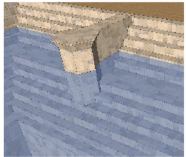
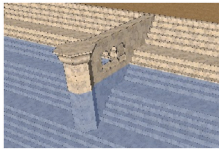
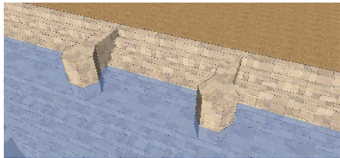
These communities' cultural and economic practices heavily rely on the availability of water, reflecting the vital role of water resources in their lives. Given the limited access to readily available water, their active engagement in community participatory water management practices is evident. In the 15th century, the advent of saints like Chaitanya Mahaprabhu and Vallabha Acharya, who were devoted to Lord Krishna, further recognized and celebrated the association of sites, including the Kunds, with Krishna. This recognition contributed to the continued development of traditional water structures in the region.

However, rapid urban sprawl and encroachment are reducing the size of these water bodies and disrupting the natural interconnected network of water flow. Catchment areas used to be covered with sacred groves or vans, which are an integral part of the ecosystem, but are now vanishing, causing soil erosion during rainfall. This causes a large quantity of silt, mud, and pollutants to get deposited on the surface of the kund, further reducing the groundwater percolation and rendering these water bodies unsuitable for use. When it comes to the development and restoration of urban water bodies, traditional techniques are usually not followed, and non-scientific approaches are adopted, around these sacred and ecologically sensitive sites. As a result, the symbiotic relationship between heritage, ecosystem, and community has been completely disrupted. This erosion of the ecological and cultural balance has significantly impacted the local population of Govardhan. The depletion of water bodies severely affects water availability for locals, especially those from marginalized communities.

“पाली पाडल नीमगांव, पानी नांय तीन गांव। “

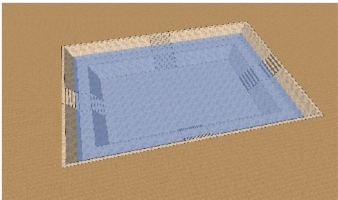
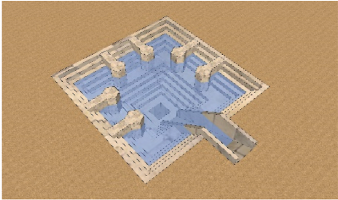
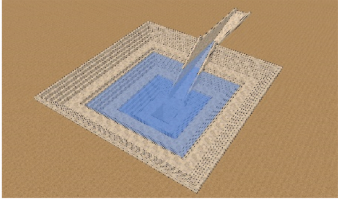
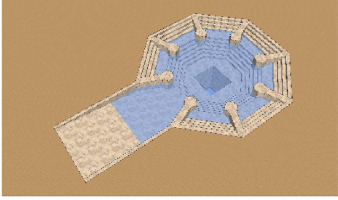
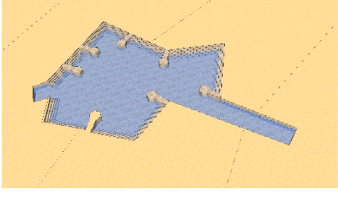
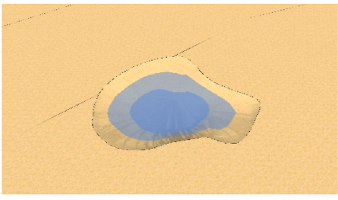
(translated as "Village named Paali Paalad and Nemgaon does not have water at all" in Brajbhasha dialect of Hindi Language), frequently

Table 1
Architectural components of Kunds.

Component	Schematic Drawing	Description
Gau Ghat Type 1		A Gau ghat is an architectural element in kunds, It serves as a means for cows and other animals to access water in these kunds. The design of gau ghat varies depending on the depth of the kund, and there are primarily three common designs observed in Govardhan. In this first cell, we'll briefly introduce the concept of gau ghat and its significance in providing water access to animals in kunds. There are three main types of gau ghat designs in Govardhan, each tailored to the specific depth of the water and the kund bed. These designs are as follows: Shallow Gau Ghat: This design is suitable for kunds with relatively shallow water and a shallower bed. It provides a gradual and easy access point for animals to reach the water source without the risk of drowning.
Gau Ghat Type 2		Deep Gau Ghat: When the kund has a deeper water level and a sufficient bed depth, a deep gau ghat is constructed. This design ensures that animals can access the deeper water while still having a secure platform, reducing the risk of accidental drowning.
Gau Ghat Type 3		Large Gau Ghat: In cases where the kund has both a significant depth of water and a large bed area, a large gau ghat is constructed. This design not only provides access to the deeper water but also accommodates a larger number of animals due to its spacious design.
Burj Type 1		Burj is a common architectural element found in kunds and is characterized by its specific features. Construction Materials: Type 1 Burj is typically constructed using bricks and sandstone. These materials are chosen for their durability and suitability for architectural needs. Size: It is relatively smaller in size compared to the other types of Burj. Commonality: This type of Burj is the most common one seen in kund architecture, reflecting its widespread use and practicality.
Burj Type 2		Type 2 Burj differs from Type 1 in several ways. Size: Type 2 Burj is larger in size when compared to Type 1. This increased size provides more surface area on top, which can be utilized for various purposes. Functionality: Due to its larger size, Type 2 Burj offers more space on top, making it suitable for a broader range of uses or activities.
Burj Type 3		Type 3 Burj represents a distinct and ornate design within Kund architecture. Ornate Design: Among the three types of Burj, Type 3 is the most ornate and visually appealing. Structural Considerations: Type 3 Burj includes specific structural considerations that make it unique in the architecture of kunds. These considerations may involve internal access features or other design elements that set it apart from the simpler Type 1 and Type 2 Burj designs. Uniqueness: Type 3 Burj stands out as a unique and sophisticated architectural feature within the context of kunds, reflecting the craftsmanship and attention to detail put into its construction.
Ghat Type 1		The common ghat design is accessible to everyone and typically consists of two burjs constructed on either side. This ghat design is easily accessible by anyone and serves as a common entry point to the kund or water body. It features two burjs (platforms or structures) on either side, providing convenient access to the water.

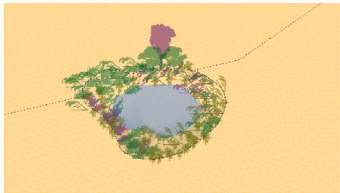
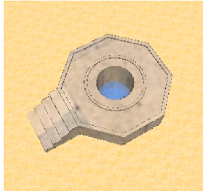
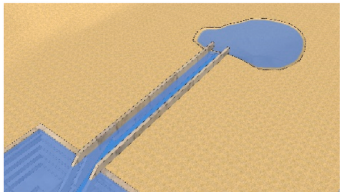
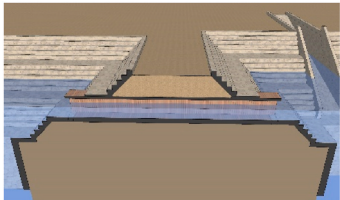
(continued on next page)

Table 1 (continued)

Component	Schematic Drawing	Description
Ghat Type 2		This ghat design is distinct from the common ghat and is characterised by an edge developed on all sides, filled with brick and lime concrete. Unlike the common ghat, this design has edges developed on all sides, providing a defined boundary to the ghat area. It is filled with brick and lime concrete, which enhances its structural integrity and longevity.
Ghat Type 3		This type of ghat design emphasises the lining of the kund (water body) all around its perimeter, ensuring structural integrity within the system. This design involves lining the entire circumference of the kund, and burjs are interspersed throughout the horizontal stretch.
Ghat Type 4		This follows the same principle as the above ghat, however on an irregular polygon-shaped kund. It follows the natural edge of a kund, without changing its natural form, on which the construction is done.
Ghat Type 5		These types of ghats are designed on the principle of a simple well, with a large depth. They depend on underground water sources and water from natural aquifers, rather than rainfall or water percolation from the surface. Their architectural details, the material used and the design are like other ghats, with an exception of the great depth that these have. Most of the kunds in the proximity of Govardhan hill are of this type.
Ghat Type 6		
Kaccha Ghat		Manually dug ghats are a type of ghat without specified built edges. They are typically constructed using brick or stone. These ghats are manually dug, and they lack defined, built edges, resulting in a more natural appearance. They are often constructed using bricks or stones, depending on local preferences and availability.

(continued on next page)

Table 1 (continued)

Component	Schematic Drawing	Description
Pokhara		These are naturally dug depressions, developed in and around the ghats. The overflowing water from the ghats goes into the <i>pokhars</i> , which essentially act like overflow sluices.
Well		Wells are primarily used to obtain water when the water quality in nearby kunds is declining or unsuitable for use. They act as a backup water source, ensuring a reliable supply even when kund water quality is compromised.
Canal		Canals are human-made structures created to transport water to various locations within the region. Most canals are open, exposed channels that carry water across the Govardhan area, facilitating the distribution of water.
Culvert		Culverts are typically located either between a Pokhara and a kund or sometimes between two kunds. They are used for the exchange or leveling of water, ensuring equitable distribution among different water bodies.
Chattri		Chattris provide shelter and a place for people to rest or seek refuge while using the kund. They are usually constructed with bricks, lime concrete, and sandstone, giving them a sturdy and durable composition. Chattris often feature arcuated structural systems, adding to their architectural charm and aesthetic appeal.

Source: Author

spoken by locals during our field study, poignantly represents the prevailing water scarcity despite the presence of a rich traditional water system. This underscores the urgency of addressing the ecological and cultural restoration of these vital water resources to ensure the sustainability of both the environment and the communities that rely on them.

These kunds were once part of the open space system of the town, but now their degradation is heavily impacting their function as a community space. As mapped in Fig. 15, Below are the linkages between TWS and the sustainable development goals and their targets:

SDG 6: Clean Water and Sanitation - TWS such as the kunds in Govardhan were traditionally a crucial source of clean water for the

local population. However, due to urban sprawl and encroachment, the size of these water bodies has been reduced and catchment areas, which were once covered with sacred groves or vans, are now vanishing. This has led to soil erosion during rainfall, causing a large quantity of silt, mud, and pollutants to get deposited on the surface of the kund. This further reduces groundwater percolation and renders these water bodies unsuitable for use. This not only affects the availability of clean water but also public health. Target 6.1: By 2030, achieve universal and equitable access to safe and affordable drinking water for all. Target 6.3: By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing the release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially

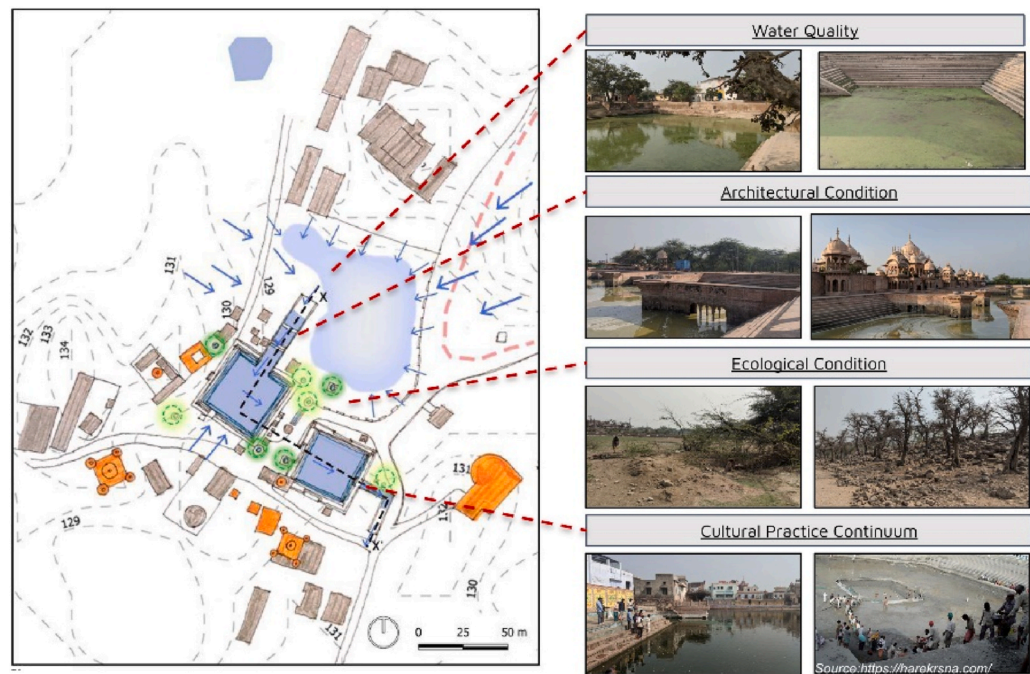


Fig. 14. Representation of a kund in a system.

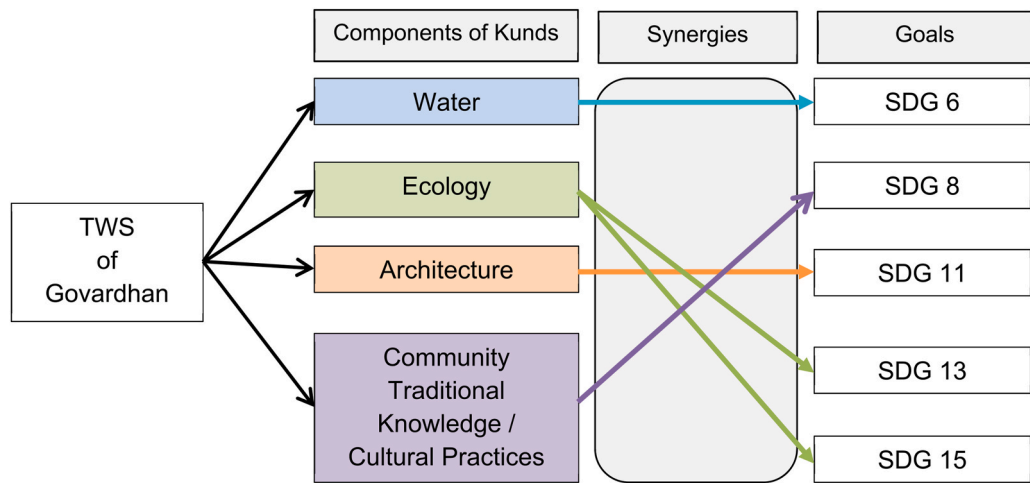


Fig. 15. A schematic representation of linkages among components of Kunds of Govardhan and SDGs.

increasing recycling and safe reuse globally.

SDG 8: Decent Work and Economic Growth - TWS in Govardhan, such as the kunds, provide employment opportunities for local communities, especially for marginalized groups such as women and children. However, their degradation has led to a decline in their function as a community space and negatively impacted the local economy. Target 8.9: By 2030, devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products.

SDG 11: Sustainable Cities and Communities - TWS in Govardhan, such as the kunds, were once an integral part of the open space system of the town, providing a community space for residents. However, due to urban sprawl and encroachment, these water bodies are now being degraded and their function as a community space has been disrupted. This negatively impacts the sustainability of the local community and overall urban planning. Target 11.6: By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management.

SDG 13: Climate Action - TWS such as the kunds in Govardhan

traditionally helped communities adapt to the impacts of climate change by providing a reliable water supply during times of drought and by reducing the risk of flooding. However, the degradation of these systems caused by urban sprawl and encroachment can now worsen the impacts of climate change on the community. Target 13.1: Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries.

SDG 15: Life on Land - TWS in Govardhan, such as the kunds, were once an integral part of the ecosystem, with catchment areas covered with sacred groves or vans. These were important for biodiversity conservation and environmental protection. However, now these sacred groves are vanishing, causing soil erosion during rainfall and leading to a decline in biodiversity and environmental protection in the region. Target 15.1: By 2020, protect and prevent the loss of biodiversity, in particular, to protect forests, wetlands, mountains and drylands, and their associated species. Target 15.2: By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase

afforestation and reforestation globally.

8. Conclusion

Water resources are an essential part of our ecosystem. By protecting this system, we are addressing a serious concern, which is the adequate availability of good quality water. Traditional knowledge of designing these water systems is helpful in the design and revival of these kunds and water systems. In the case of Govardhan, the topography of the landscape is such that it allows rainwater to be captured in shallow depressions with large catchment areas. Religious association with these water bodies has led to multiple initiatives for their improvement. Religious association with these water bodies has led to architectural developments, to further utilise these water bodies, kunds were built all around the parikrama of Govardhan.

Indigenous communities, such as the Jats, Gujars, Ahirs, and Abhiras, have deep-rooted ties to the Braj region, including Govardhan. These communities, with their diverse subcastes and unique cultural identities, have played pivotal roles in shaping the agricultural landscape and the broader socio-economic framework of Braj. A critical aspect of these communities' lives is their profound reliance on water resources. Given the scarcity of readily available water in the region, their active involvement in community-based water management practices becomes strikingly evident. Kunds have been serving the water requirement of the residents and pilgrims for generations and have also played a key role in the development of the cultural landscape of Govardhan. Regular protection and proper management are required to keep the water free from pollutants and ensure an adequate quantity of water, which is essential to maintain the ecosystem balance. These structures also served as rainwater recharge zones. They have evolved using the age-old wisdom and knowledge of geography, developed over hundreds of years. These kunds are a significant piece of built component, as it is a signature of a long lasted, rooted, and culturally thriving environments, as well are in confluence with the natural environment.

However, the rapid urban sprawl and encroachment pose imminent threats to these invaluable water bodies. The disappearance of sacred groves and the resulting soil erosion during rainfall contribute to the accumulation of silt, mud, and pollutants in the Kunds. This jeopardizes groundwater percolation and renders these water systems unsuitable for their intended purposes.

This degradation of TWS has significant implications for the achievement of several SDGs, including SDG 6 (Clean Water and Sanitation), SDG 8 (Decent Work and Economic Growth), SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land). To address this multifaceted challenge and ensure the sustainable development of the Govardhan region, it is imperative to prioritize the protection, restoration, and holistic management of these traditional water systems. Recognising the integral role of Indigenous communities in this endeavour is paramount. Their ancestral knowledge and practices hold the key to preserving the delicate balance between nature, culture, and human livelihoods. Thus, any efforts to safeguard these water resources must be inclusive, empowering these communities, and acknowledging their rights and contributions.

CRedit authorship contribution statement

Aman Sharma: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Shiva Ji:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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